

A Technical Introduction to Graph attention network

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Section 1

Graph neural networks are one of the main building blocks of AlphaFold, an artificial intelligence program developed by Google's DeepMind for solving the protein folding problem in biology. AlphaFold achieved first place in several CASP competitions.

Section 2

Message passing layers are permutation-equivariant layers mapping a graph into an updated representation of the same graph. Formally, they can be expressed as message passing neural networks (MPNNs). Let $G = (V, E)$ $\{\displaystyle G=(V,E)\}$ be a graph, where V $\{\displaystyle V\}$ is the node set and E $\{\displaystyle E\}$ is the edge set. Let N_u $\{\displaystyle N_{\{u\}}\}$ be the neighbourhood of some node $u \in V$ $\{\displaystyle u \in V\}$. Additionally, let $\mathbf{x}_u$ $\{\displaystyle \mathbf{x}_{\{u\}}\}$ be the features of node $u \in V$ $\{\displaystyle u \in V\}$, and $\mathbf{e}_{uv}$ $\{\displaystyle \mathbf{e}_{\{uv\}}\}$ be the features of edge $(u, v) \in E$ $\{\displaystyle (u,v) \in E\}$. An MPNN layer can be expressed as follows:

Section 3

Graph neural networks (GNNs) are specialized artificial neural networks that are designed for tasks whose inputs are graphs. One prominent example is molecular drug design. Each input sample is a graph representation of a molecule, where atoms form the nodes and chemical bonds between atoms form the edges. In addition to the graph representation, the input also includes known chemical properties for each of the atoms. Dataset samples may thus differ in length, reflecting the varying numbers of atoms in molecules, and the varying number of bonds between them. The task is to predict the efficacy of a given molecule for a specific medical application, such as eliminating *E. coli* bacteria. The key design element of GNNs is the use of pairwise message passing, such that graph nodes iteratively update their representations by exchanging information with their neighbors. Several GNN architectures have been proposed, which implement different flavors of message passing, started by recursive or convolutional constructive approaches. As of 2022, it is an open question whether it is possible to define GNN architectures "going beyond" message passing, or instead every GNN can be built on message passing over suitably defined graphs.

Section 4

Permutation equivariant: a permutation equivariant layer maps a representation of a graph into an updated representation of the same graph. In the literature, permutation equivariant layers are implemented via pairwise message passing between graph nodes. Intuitively, in a message passing layer, nodes update their representations by aggregating the messages received from their immediate neighbours. As such, each message passing layer increases the receptive field of the GNN by one hop. Local pooling: a local pooling layer coarsens the graph via downsampling. Local pooling is used to increase the receptive field of a GNN, in a similar fashion to pooling layers in convolutional neural networks. Examples include k-nearest neighbours pooling, top-k pooling, and self-attention pooling. Global pooling: a global pooling layer, also known as readout layer, provides fixed-size representation of the whole graph. The global pooling layer must be permutation invariant, such that permutations in the ordering of graph nodes and edges do not alter the final output. Examples include element-wise sum, mean or maximum. It has been demonstrated that GNNs cannot be more expressive than the Weisfeiler–Leman Graph Isomorphism Test. In practice, this means that there exist different graph structures (e.g., molecules with the same atoms but different bonds) that cannot be distinguished by GNNs. More powerful GNNs operating on higher-dimension geometries such as simplicial complexes can be designed. As of 2022, whether or not future architectures will overcome the message passing primitive is an open research question.

Section 5

Gated graph sequence neural network The gated graph sequence neural network (GGS-NN) was introduced by Yujia Li et al. in 2015. The GGS-NN extends the GNN formulation by Scarselli et al. to output sequences. The message passing framework is implemented as an update rule to a gated recurrent unit (GRU) cell. A GGS-NN can be expressed as follows:

Section 6

where $i = \text{top}_k(\mathbf{y})$ $\{\displaystyle \mathbf{i} = \text{top}_k(\mathbf{y})\}$ is the subset of nodes with the top-k highest projection scores, $\odot$ $\{\displaystyle \odot\}$ denotes element-wise matrix multiplication. The self-attention pooling layer can be seen as an extension of the top-k pooling layer. Differently from top-k pooling, the self-attention scores computed in self-attention pooling account both for the graph features and the graph topology.

Section 7

$H = \sigma(D \sim -1/2 A \sim -1/2 X \Theta)$ $\{\displaystyle H = \sigma(D \sim -1/2 A \sim -1/2 X \Theta)\}$ $= \sum \left(\tilde{\mathbf{D}}^{\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{\frac{1}{2}} \right) \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{\frac{1}{2}}$

$$\{X\} \mathbf{\Theta}$$

Section 8

where $\mathbf{p}$ is a learnable projection vector. The projection vector $\mathbf{p}$ computes a scalar projection value for each graph node. The top-k pooling layer can then be formalised as follows:

Section 9

Heterophilic Graph Learning Homophily principle, i.e., nodes with the same labels or similar attributes are more likely to be connected, has been commonly believed to be the main reason for the superiority of Graph Neural Networks (GNNs) over traditional Neural Networks (NNs) on graph-structured data, especially on node-level tasks. However, recent work has identified a non-trivial set of datasets where GNN's performance compared to the NN's is not satisfactory. Heterophily, i.e., low homophily, has been considered the main cause of this empirical observation. People have begun to revisit and re-evaluate most existing graph models in the heterophily scenario across various kinds of graphs, e.g., heterogeneous graphs, temporal graphs and hypergraphs. Moreover, numerous graph-related applications are found to be closely related to the heterophily problem, e.g. graph fraud/anomaly detection, graph adversarial attacks and robustness, privacy, federated learning and point cloud segmentation, graph clustering, recommender systems, generative models, link prediction, graph classification and coloring, etc. In the past few years, considerable effort has been devoted to studying and addressing the heterophily issue in graph learning.

Section 10

Local pooling layers Local pooling layers coarsen the graph via downsampling. Subsequently, several learnable local pooling strategies that have been proposed are presented. For each case, the input is the initial graph represented by a matrix X of node features, and the graph adjacency matrix A . The output is the new matrix X' of node features, and the new graph adjacency matrix A' .